

SOLARIS ADMINISTRATION FOR ORACLE DBA

OS Basics | File Systems | User Management | ZFS | Zones | Storage | Networking

With Real-Time Examples, Commands & Interview Tips

April 2026 — Intermediate Level

Introduction

Oracle databases in large enterprises — especially banking, finance, telecom, and government — run predominantly on Solaris/SPARC infrastructure. An Oracle DBA who understands the Solaris operating system is significantly more valuable than one who only knows Linux.

This handbook is designed for you — an intermediate Solaris user who has worked with OS basics, file systems, and user management, and now wants to go deeper and tie Solaris skills directly to Oracle DBA work.

By the end of this guide you will be confident in:

- Solaris OS internals and performance monitoring
- File systems — UFS, ZFS — and how they relate to Oracle datafiles
- User and group management including Oracle-specific OS setup
- Storage management — ZFS pools, SVM, Veritas VxVM
- Solaris Zones — running Oracle inside containers
- Networking — relevant to Oracle listener and connectivity
- SMF — Service Management Facility for Oracle service control
- Interview-level command knowledge across all topics

PART 1

Solaris OS Fundamentals — Going Deeper

Chapter 1: Solaris Architecture & Versions

Understanding which version of Solaris you are on is the first step, because commands and tools differ significantly between versions.

1.1 Solaris Versions You Will Encounter

Version	Key Features	Where You'll See It
Solaris 8 / 9	Legacy — SVM, UFS, /etc/system kernel params	Very old prod systems, rare
Solaris 10	Zones, SMF, DTrace, ZFS, RBAC, iSCSI	Still very common in enterprises
Solaris 11 / 11.4	IPS packaging, unified archives, immutable zones, full ZFS	Modern Oracle shops, ODA
Oracle Solaris 11.4 SRU	Latest — monthly SRUs (Support Repository Updates)	New deployments

1.2 Checking Your Solaris Version

```
# Check Solaris version
uname -a
cat /etc/release

# Solaris 11 — check SRU level
pkg info entire | grep -i branch

# Check hardware architecture
uname -p          # sparc or i386
isainfo -kv       # kernel instruction set
prtconf | head    # hardware config, total memory
```

NOTE: SPARC systems use Big-Endian byte order. If you move Oracle datafiles between SPARC and x86, they are not directly compatible.

1.3 Solaris Boot Process

Understanding the boot process helps you diagnose startup issues — very relevant when Oracle doesn't start after a reboot.

Stage	What Happens
POST	Power-On Self Test — hardware check by OpenBoot PROM (OBP)
OBP / GRUB	Boot loader selects kernel — OpenBoot on SPARC, GRUB on x86
Kernel loads	/platform/.../kernel/unix loaded into memory
init / SMF	init starts SMF (Service Management Facility) — replaces old rc scripts
Milestone reached	SMF brings services online in dependency order — includes Oracle if configured

```
# Boot related commands
who -b                # when was system last booted
last reboot           # reboot history
uptime                # how long system has been up

# From OBP prompt (SPARC) — only at console
ok> boot -s           # boot into single-user mode (maintenance)
ok> boot -r           # reconfigure boot (rescan devices)
```

Chapter 2: Solaris File System Administration

As an Oracle DBA on Solaris, you constantly work with file systems — placing Oracle software, datafiles, archive logs, and backups. Knowing file systems deeply is essential.

2.1 Key Solaris File Systems

File System	Type	Oracle Use Case
UFS	Unix File System — traditional Solaris FS	Oracle SW on older Solaris 10 systems
ZFS	Zettabyte File System — modern, pooled	Oracle datafiles, FRA, archive logs on Sol 11
tmpfs	Memory-based temporary FS	/tmp — Oracle temp space, careful not to confuse with TEMP tablespace
procfs	Virtual — process information	/proc — Oracle process diagnostics
NFS	Network File System	Shared Oracle homes in clustered setups (with care)
VxFS	Veritas File System	Oracle datafiles in Veritas environments

2.2 Disk and Partition Management

Viewing Disks

```
# List all disks
format                                # interactive disk utility — shows all disks
iostat -En                           # disk info including size and errors
cfgadm -al                           # all devices including unattached

# Solaris 11 — list disks
diskinfo -l                          # detailed disk listing
format -e                            # expert mode format
```

Partitioning — fdisk and format

```
# View partition table of a disk
prtvtoc /dev/rdisk/c0t0d0s2          # VTOC partition table (SPARC)
fdisk /dev/rdisk/c0t1d0p0            # x86 fdisk partition table

# Create a new slice using format
format /dev/rdisk/c0t2d0
partition                            # enter partition menu
3                                    # select slice 3
0                                    # tag = unassigned
wm                                   # permissions
```

```
0                # start cylinder
10gb             # size
label           # write label
quit
```

NOTE: In modern Solaris 11 with ZFS, you rarely partition disks manually. You hand entire disks to ZFS pools. Partitioning is mainly needed on Solaris 10 with UFS or SVM.

2.3 UFS — Unix File System

UFS is the traditional Solaris file system. You may still encounter it on Solaris 10 systems especially for the root file system and Oracle ORACLE_HOME.

Creating and Mounting UFS

```
# Create UFS file system on a slice
newfs /dev/rdisk/c0t2d0s3

# Mount UFS file system
mount -F ufs /dev/dsk/c0t2d0s3 /u01

# Make it permanent — add to /etc/vfstab
/dev/dsk/c0t2d0s3 /dev/rdisk/c0t2d0s3 /u01 ufs 2 yes -

# Check UFS usage
df -h /u01
df -k                # all mounted file systems in KB
```

UFS Maintenance

```
# Check and repair UFS (must unmount first)
umount /u01
fsck -F ufs /dev/rdisk/c0t2d0s3

# Check UFS fragmentation
fstyp -v /dev/rdisk/c0t2d0s3 | grep -i frag

# Grow UFS (if underlying volume grew)
growfs -s <new_size_sectors> /dev/rdisk/c0t2d0s3
```

2.4 ZFS — Zettabyte File System (Critical for Modern Oracle DBAs)

ZFS is Oracle's most advanced file system and is the standard on Solaris 11. It combines volume management, RAID, and file system into one. For Oracle DBAs it eliminates the need for separate LVM tools.

ZFS Key Concepts

Concept	Meaning	Oracle Analogy
zpool	Storage pool — collection of physical disks	Like an ASM Disk Group
Dataset	File system, volume, or snapshot within a zpool	Like an ASM File
Vdev	Virtual device — a disk or mirror/raidz group within a pool	Like an ASM Disk
Snapshot	Read-only point-in-time copy of a dataset	Like a Flashback snapshot
Clone	Writable copy of a snapshot	Like an EBS clone from snapshot
Quota	Maximum size a dataset can grow to	Like tablespace max size
Reservation	Guaranteed space for a dataset	Like tablespace initial size

ZFS Pool Operations

```
# Create a simple pool (single disk — dev/test only)
zpool create orapool /dev/dsk/c0t2d0

# Create a mirrored pool (production standard)
zpool create orapool mirror /dev/dsk/c0t2d0 /dev/dsk/c0t3d0

# Create RAIDZ pool (like RAID-5, tolerates 1 disk failure)
zpool create orapool raidz /dev/dsk/c0t2d0 /dev/dsk/c0t3d0 /dev/dsk/c0t4d0

# Create RAIDZ2 pool (tolerates 2 disk failures — production recommended)
zpool create orapool raidz2 /dev/dsk/c0t2d0 /dev/dsk/c0t3d0 /dev/dsk/c0t4d0 /dev/dsk/c0t5d0

# List pools
zpool list
zpool list -v orapool           # verbose — shows vdevs

# Pool status and health
zpool status
zpool status orapool

# Pool I/O statistics
zpool iostat 5                 # refresh every 5 seconds
zpool iostat -v orapool 5      # per-vdev I/O stats

# Add a disk to existing pool (expand storage)
zpool add orapool /dev/dsk/c0t6d0

# Add a spare disk
zpool add orapool spare /dev/dsk/c0t7d0
```

```
# Scrub pool (verify data integrity – like RMAN VALIDATE)
zpool scrub orapool
zpool status orapool          # check scrub progress
```

ZFS Dataset Operations

```
# Create datasets for Oracle
zfs create orapool/oracle_base  # Oracle base
zfs create orapool/oracle_home  # Oracle Home
zfs create orapool/oradata      # Oracle datafiles
zfs create orapool/fra          # Fast Recovery Area
zfs create orapool/arch         # Archive logs

# Set mountpoint
zfs set mountpoint=/u01 orapool/oracle_base
zfs set mountpoint=/u01/app/oracle/product/19c orapool/oracle_home
zfs set mountpoint=/u01/oradata orapool/oradata

# List datasets
zfs list
zfs list -r orapool          # recursive list
```

ZFS Properties — Oracle Tuning

```
# CRITICAL: Set recordsize for Oracle datafiles (match Oracle block size)
zfs set recordsize=8k orapool/oradata    # for 8KB Oracle block size

# Disable atime (access time updates) – reduces unnecessary I/O
zfs set atime=off orapool/oradata

# Set compression (saves space, minimal CPU overhead)
zfs set compression=lz4 orapool/oradata

# For Oracle redo logs – sync writes critical
zfs set sync=always orapool/redologs     # guarantees redo safety

# For archive logs – async is fine (already committed)
zfs set sync=disabled orapool/arch

# Set quota (max size for a dataset)
zfs set quota=500G orapool/oradata

# Set reservation (guaranteed space)
zfs set reservation=100G orapool/oradata

# View all properties of a dataset
zfs get all orapool/oradata
zfs get recordsize,atime,compression,sync orapool/oradata
```


ZFS Snapshots — Powerful for Oracle DBAs

ZFS snapshots are instant, space-efficient, and perfect for quick database backups or cloning test environments.

```
# Create a snapshot (instant – no data copied yet)
zfs snapshot orapool/oradata@before_patch_20260420

# List snapshots
zfs list -t snapshot

# Rollback to snapshot (database must be down)
zfs rollback orapool/oradata@before_patch_20260420

# Clone a snapshot (for DEV/TEST environment)
zfs clone orapool/oradata@before_patch_20260420 orapool/oradata_dev

# Delete snapshot
zfs destroy orapool/oradata@before_patch_20260420
```

TIP: Real-time scenario: Before applying an Oracle patch, take a ZFS snapshot of the Oracle Home and datafiles dataset. If the patch fails, rollback takes seconds — much faster than an RMAN restore.

2.5 /etc/vfstab — File System Mount Table

This is the Solaris equivalent of Linux /etc/fstab. It defines what gets mounted at boot.

Field	Meaning	Example
device to mount	Block device path	/dev/dsk/c0t2d0s3
device to fsck	Raw device for fsck	/dev/rdisk/c0t2d0s3
mount point	Directory to mount at	/u01
FS type	File system type	ufs or zfs
fsck pass	fsck check order (0=skip)	2
mount at boot	yes or no	yes
mount options	Mount flags	- (dash means default)

```
# Example /etc/vfstab entries
/dev/dsk/c0t2d0s3 /dev/rdisk/c0t2d0s3 /u01      ufs  2  yes  logging
-                -                /orapool  zfs  -   yes  -

# Mount all entries in vfstab
mountall
```

```
# Unmount all non-system file systems
umountall
```

Chapter 3: User and Group Management

User and group management on Solaris is something you already know. Let's go deeper — especially around Oracle-specific OS user setup, RBAC, and security profiles.

3.1 Oracle OS User Setup on Solaris

Before installing Oracle on Solaris, you must create specific OS users and groups. This is different from Linux — Solaris has stricter defaults.

Creating Oracle Groups and User

```
# Create required Oracle groups
groupadd -g 100 oinstall          # Oracle inventory group
groupadd -g 101 dba              # SYSDBA privilege group
groupadd -g 102 oper             # SYSOPER privilege group
groupadd -g 103 backupdba        # SYSBACKUP privilege group

# Create oracle user
useradd -u 200 -g oinstall -G dba,oper,backupdba -d /export/home/oracle -m -s
/bin/bash oracle

# Set password
passwd oracle

# Verify user creation
id oracle
finger oracle
```

Oracle User Shell Profile (.bash_profile or .profile)

```
# /export/home/oracle/.bash_profile
export ORACLE_BASE=/u01/app/oracle
export ORACLE_HOME=/u01/app/oracle/product/19.0.0/dbhome_1
export ORACLE_SID=ORCL
export PATH=$ORACLE_HOME/bin:/usr/bin:/usr/sbin:/usr/local/bin:$PATH
export LD_LIBRARY_PATH=$ORACLE_HOME/lib:/usr/lib
export CLASSPATH=$ORACLE_HOME/jlib:$ORACLE_HOME/rdbms/jlib
export NLS_DATE_FORMAT='DD-MON-YYYY HH24:MI:SS'
export TMP=/tmp
export TMPDIR=/tmp

# Solaris-specific: For EBS also set
export TWO_TASK=ORCL
export ORACLE_TERM=vt100
```

3.2 Solaris Kernel Parameters for Oracle

On Solaris 10, Oracle kernel parameters are set via `/etc/system` and projects. On Solaris 11 they are managed via projects.

Solaris 10 — `/etc/system`

```
# Add to /etc/system for Oracle shared memory
set shmsys:shminfo_shmmax=4294967295
set shmsys:shminfo_shmseg=256
set semsys:seminfo_semmni=256
set semsys:seminfo_semmns=32768
set semsys:seminfo_semopm=100
set semsys:seminfo_semvms=32767
```

NOTE: Changes to `/etc/system` require a reboot to take effect on Solaris 10.

Solaris 10/11 — Projects (Preferred Method)

```
# Create a project for Oracle
projadd -c 'Oracle Database' -U oracle -G dba -K 'project.max-shm-
memory=(privileged,8589934592,deny)' oracle

# View projects
projects -l
projects oracle                # projects for oracle user

# Check resource controls
prctl -i project oracle

# Modify project resource — shared memory
projmod -s -K 'project.max-shm-memory=(privileged,17179869184,deny)' oracle

# Set semaphore limits
projmod -s -K 'project.max-sem-ids=(privileged,256,deny)' oracle
projmod -s -K 'process.max-sem-nsems=(privileged,256,deny)' oracle
```

3.3 User Administration Commands

Managing Users

```
# Add user
useradd -u 205 -g oinstall -G dba -d /export/home/appsdba -m -s /bin/bash appsdba

# Modify user — add to another group
usermod -G dba,oper oracle
```

```
# Lock a user account
passwd -l username

# Unlock a user account
passwd -u username

# Delete user
userdel -r username           # -r removes home directory

# Check user info
getent passwd oracle          # check user in /etc/passwd
id oracle                     # UID, GID, groups
who                            # who is logged in
last oracle                    # login history
finger oracle                  # detailed user info
```

Key User Files on Solaris

File	Purpose
/etc/passwd	User accounts — username, UID, GID, home, shell
/etc/shadow	Encrypted passwords and expiry settings
/etc/group	Group definitions
/etc/user_attr	RBAC attributes for users (Solaris-specific)
/etc/security/auth_attr	Authorization definitions
/etc/security/exec_attr	Execution profiles

3.4 RBAC — Role-Based Access Control

RBAC is a Solaris feature that allows fine-grained privilege delegation without giving full root access. Very useful for Oracle DBAs who need certain root-level tasks.

```
# List available profiles
profiles

# Assign a profile to a user — e.g., give oracle user 'System Administrator' profile
usermod -P 'System Administrator' oracle

# Run a command with a profile (pfexec — like sudo but RBAC-aware)
pfexec /usr/sbin/mount /dev/dsk/c0t2d0s3 /u01

# Grant specific authorization
usermod -A solaris.system.shutdown oracle

# Check user authorizations
```

```
auths oracle
```

TIP: RBAC is preferred over direct sudo in Solaris environments. It provides better audit trails and is more granular.

3.5 sudo on Solaris

```
# Check if sudo is installed
which sudo

# Edit sudoers file (always use visudo)
visudo

# Example sudoers entry for oracle user
oracle ALL=(root) /usr/sbin/mount, /usr/sbin/umount, /usr/bin/pkill

# Verify sudo access
sudo -l -U oracle
```

Chapter 4: Solaris Performance Monitoring

Performance monitoring on Solaris is a daily task — especially when Oracle is slow and you need to determine if the problem is at the OS level (CPU, memory, I/O) or at the Oracle level.

4.1 CPU Monitoring

```
# Basic CPU snapshot
uptime                                # load average for 1, 5, 15 mins

# vmstat - CPU, memory, I/O summary
vmstat 5                             # report every 5 seconds
vmstat 5 10                          # 10 reports, every 5 seconds
```

vmstat Column	Meaning	Oracle Concern if High
r	Processes in run queue	CPU saturation — Oracle processes waiting for CPU
b	Processes blocked (waiting for I/O)	I/O bottleneck — Oracle I/O heavy
us	User CPU %	Oracle SQL, PL/SQL consuming CPU
sy	System CPU %	Kernel/OS activity — if very high, OS issue
id	Idle CPU %	If near 0, CPU is saturated
wa	I/O wait %	Disk I/O bottleneck — Oracle reading/writing heavily

```
# prstat - Solaris equivalent of Linux 'top'
prstat                                # all processes, refreshes every 5s
prstat -a                             # per-user summary at bottom
prstat -p <PID>                       # monitor specific Oracle process
prstat -u oracle                      # all Oracle processes
prstat 1 20                          # 20 refreshes, every 1 second

# mpstat - per-CPU statistics (for multi-CPU SPARC servers)
mpstat 5

# psrinfo - CPU info
psrinfo -v                            # detailed CPU info
psrinfo | wc -l                       # count virtual CPUs
```

4.2 Memory Monitoring

```
# prtconf - total physical memory
prtconf | grep Memory

# vmstat swap stats
```

```
swap -s                # swap space summary
swap -l                # list swap devices

# mdb - Solaris debugger (advanced) - check kernel memory
echo '::memstat' | mdb -k

# sar - System Activity Reporter
sar -r 5 10            # memory: free pages every 5s, 10 times
sar -g 5 10            # page-out activity
```

NOTE: Oracle SGA is locked into memory (with LOCK_SGA=TRUE or AMM). If system has insufficient RAM, Oracle competes with OS for memory causing severe paging. Always ensure system RAM > SGA + PGA + OS overhead.

4.3 Disk I/O Monitoring

```
# iostat - disk I/O statistics (most used by Oracle DBAs)
iostat -xnzc 5          # extended stats, device names, skip idle
```

iostat Column	Meaning	Concern Level
r/s	Reads per second	Baseline depends on workload
w/s	Writes per second	High during bulk inserts or RMAN
kr/s	Kilobytes read per second	Monitor for I/O saturation
kw/s	Kilobytes written per second	High during checkpoint, redo flush
%b	% time disk was busy	Above 80% = bottleneck
asvc_t	Average service time in ms	Above 20ms for SAN = investigate
wait	Average queue length	Above 1 = I/O queuing

```
# Monitor specific disk
iostat -xn 5 | grep c0t0d0

# sar for I/O
sar -d 5 10          # disk activity every 5s
```

4.4 Network Monitoring

```
# netstat - network connections
netstat -an           # all connections with addresses
netstat -an | grep 1521 # Oracle listener connections
netstat -s             # protocol statistics

# Check Oracle listener connections
netstat -an | grep LISTEN | grep 1521
```



```
# nicstat - NIC statistics (if installed)
nicstat 5
```

4.5 DTrace — Solaris Superpower

DTrace is Solaris's unique dynamic tracing framework. It lets you trace ANYTHING happening in the system — kernel, applications, system calls — with zero overhead when not in use.

```
# Count syscalls by process name
dtrace -n 'syscall:::entry { @[execname] = count(); }'

# Trace Oracle process file I/O
dtrace -n 'syscall::read:entry /pid == <oracle_pid>/ { @[fds[arg0].fi_name] =
sum(arg2); }'

# Monitor Oracle SQL*Net read/write calls
dtrace -n 'syscall::read:entry,syscall::write:entry /execname == "oracle"/ {
@[probecount] = count(); }'

# Trace slow file reads (> 10ms)
dtrace -n 'syscall::read:entry { self->t = timestamp; } syscall::read:return /self->t
&& (timestamp-self->t)/1000000 > 10/ { printf("%s: %d ms", execname, (timestamp-self-
>t)/1000000); self->t = 0; }'
```

TIP: DTrace is one of the biggest advantages Solaris has over Linux. Even basic DTrace knowledge sets you apart in interviews. Practice a few one-liners.

Chapter 5: SMF — Service Management Facility

SMF replaced the old `/etc/rc.d` init scripts in Solaris 10+. It manages all system services including Oracle — if you configure Oracle as an SMF service, it auto-starts on boot and restarts on failure.

5.1 SMF Key Concepts

Concept	Meaning
FMRI	Fault Management Resource Identifier — unique name for each service e.g. <code>svc:/network/ssh:default</code>
Milestone	Special service that groups others — like multi-user-server milestone
Manifest	XML file defining a service — lives in <code>/lib/svc/manifest/</code>
Method	Script that SMF calls to start/stop the service
Restarter	SMF component that monitors and restarts failed services

5.2 Core SMF Commands

```
# List all services and their state
svcs -a                                # all services
svcs -a | grep oracle                  # Oracle-related services

# Service states
online      -- running normally
offline     -- not running (dependencies not met)
disabled    -- administratively disabled
maintenance -- failed, needs attention

# Enable a service (starts now + survives reboots)
svcadm enable svc:/application/oracle/database:ORCL

# Disable a service
svcadm disable svc:/application/oracle/database:ORCL

# Restart a service
svcadm restart svc:/network/ssh:default

# Refresh (re-read config without restart)
svcadm refresh svc:/network/dns/client:default

# Clear a service in maintenance state
svcadm clear svc:/application/oracle/database:ORCL

# Check why a service is in maintenance
```

```
svcs -x svc:/application/oracle/database:ORCL

# View service log
svcs -l svc:/application/oracle/database:ORCL    # find log path
tail -50 /var/svc/log/application-oracle-database:ORCL.log
```

5.3 Configuring Oracle as SMF Service

Configuring Oracle as an SMF service means Oracle auto-starts on reboot and auto-recovers from crashes — no manual intervention needed.

Create Oracle SMF Manifest

```
# Create method script: /lib/svc/method/oracle-db
#!/bin/bash
. /export/home/oracle/.bash_profile
case $1 in
    start)
        su - oracle -c 'sqlplus / as sysdba <<EOF
        STARTUP;
        EXIT;
        EOF'
        su - oracle -c 'lsnrctl start'
        ;;
    stop)
        su - oracle -c 'sqlplus / as sysdba <<EOF
        SHUTDOWN IMMEDIATE;
        EXIT;
        EOF'
        su - oracle -c 'lsnrctl stop'
        ;;
esac

chmod +x /lib/svc/method/oracle-db

# Import manifest
svccfg import /lib/svc/manifest/application/oracle-db.xml

# Enable the service
svcadm enable svc:/application/oracle/db:ORCL
```

Chapter 6: Storage Management — SVM and Veritas

In Solaris environments you will encounter two major volume managers besides ZFS: Solaris Volume Manager (SVM) and Veritas Volume Manager (VxVM). Both are used to manage Oracle storage on Solaris 10 systems.

6.1 Solaris Volume Manager (SVM)

SVM is the free, built-in volume manager for Solaris 10. It provides mirroring, striping, and RAID-5 without additional cost.

SVM Key Components

Component	Description	Oracle Use
metadevice	Virtual disk device created by SVM	/dev/md/dsk/d10 used as Oracle datafile device
metaset	Named set of metadevices (for clustering)	Used in Sun Cluster with Oracle
submirror	One copy in a mirror	Each submirror = physical disk slice
state database	SVM metadata — must have at least 3 replicas	Critical — loss means SVM fails

SVM Setup and Commands

```
# Initialize state database replicas (do this first)
metadb -a -f -c 3 c0t0d0s7 c0t1d0s7 c0t2d0s7

# Create a simple concat metadevice
metainit d10 1 1 c0t2d0s3

# Create a stripe (2 disks striped)
metainit d20 1 2 c0t2d0s3 c0t3d0s3

# Create a mirror
metainit d30 -m c0t2d0s3          # create with first submirror
metattach d30 c0t3d0s3          # attach second submirror (sync begins)

# Check mirror sync status
metastat d30

# List all metadevices
metastat
metastat -p                      # parseable output
```

```
# Grow a metadvice (after adding space)
metattach dl0 c0t4d0s3          # add new slice to concat

# Replace a failed disk in a mirror
metareplace d30 c0t3d0s3 c0t5d0s3 # replace failed submirror slice
```

6.2 Veritas Volume Manager (VxVM)

Veritas VxVM is a commercial volume manager widely used in large enterprises. You will encounter it frequently in bank and telecom Oracle environments.

VxVM Concepts

VxVM Term	Meaning	Equivalent in SVM/ZFS
Disk Group	Named collection of physical disks	SVM metaset / ZFS zpool
Volume	Virtual disk created from disk group	SVM metadvice / ZFS dataset
Plex	One complete copy of a volume	SVM submirror / ZFS vdev
Subdisk	Portion of a physical disk	SVM slice
VxFS	Veritas File System — created on VxVM volumes	UFS / ZFS

VxVM Commands

```
# Check VxVM disk group
vxdg list          # list all disk groups
vxdg -g oradg list # details of oradg disk group

# List volumes
vxprint -g oradg -ht # volumes in oradg group

# Check volume status
vxprint -g oradg -v  # volume status

# Create a volume (2GB, mirrored)
vxassist -g oradg make oravol01 2g layout=mirror

# Grow a volume
vxassist -g oradg growby oravol01 1g

# Check I/O statistics on VxVM volumes
vxstat -g oradg -d          # per-disk stats
vxstat -g oradg             # per-volume stats

# Start disk group (after reboot or failover)
vxdg import oradg
```

```
# Stop disk group
vxdg deport oradg

# Replace a failed disk in VxVM
vxdisksetup -i /dev/rdisk/c0t5d0
vxdg -g oradg adddisk newdisk=c0t5d0
vxevac c0t3d0                # evacuate failed disk
vxdg -g oradg rmdisk c0t3d0
```

Chapter 7: Solaris Zones — Oracle in Containers

Solaris Zones (also called Solaris Containers) provide OS-level virtualization. Multiple isolated environments run on a single Solaris system. Oracle databases commonly run inside non-global zones for isolation and resource control.

7.1 Zone Types

Zone Type	Description	Oracle Use
Global Zone	The main OS — has full hardware access	System administration, zone management
Non-Global Zone (sparse)	Shares some OS directories with global zone	Older Oracle setups on Solaris 10
Non-Global Zone (whole root)	Has its own complete copy of OS	Oracle on Solaris 10 — full isolation
Solaris 11 Zone	Native zone — always whole root on Solaris 11	Oracle 19c on Solaris 11
Kernel Zone	Full virtualization — own kernel	Heavy isolation, Oracle for critical DBs
Immutable Zone	Read-only OS — enhanced security	Oracle with security hardening

7.2 Zone Administration Commands

```
# List all zones
zoneadm list -v                # verbose - all zones with state

# Zone states
running    -- zone is up and running
installed  -- zone configured and installed, but not running
configured -- zone only configured, not yet installed
ready      -- zone kernel loaded, not yet booted (transient)

# Boot a zone
zoneadm -z orazone boot

# Halt a zone (like shutdown immediate)
zoneadm -z orazone halt

# Graceful shutdown of a zone
zoneadm -z orazone shutdown

# Reboot a zone
zoneadm -z orazone reboot
```

```
# Login to a zone console
zlogin -C orazone           # console — even if zone won't boot

# Login to zone shell
zlogin orazone              # interactive shell inside zone

# Run a command inside a zone from global zone
zlogin orazone 'ps -ef | grep oracle'
zlogin orazone 'lsnrctl status'
```

7.3 Creating a Zone for Oracle

```
# Step 1: Create zone configuration
zonecfg -z orazone
zonecfg:orazone> create
zonecfg:orazone> set zonepath=/zones/orazone
zonecfg:orazone> set autoboot=true
zonecfg:orazone> set ip-type=exclusive
zonecfg:orazone> add net
zonecfg:orazone:net> set physical=net1
zonecfg:orazone:net> set address=192.168.1.100/24
zonecfg:orazone:net> end
# Add Oracle data directory from global zone
zonecfg:orazone> add fs
zonecfg:orazone:fs> set dir=/u01
zonecfg:orazone:fs> set special=/orapool/oracle_home
zonecfg:orazone:fs> set type=zfs
zonecfg:orazone:fs> end
# Set CPU and memory caps
zonecfg:orazone> set cpu-shares=512
zonecfg:orazone> add capped-memory
zonecfg:orazone:capped-memory> set physical=16G
zonecfg:orazone:capped-memory> end
zonecfg:orazone> verify
zonecfg:orazone> commit
zonecfg:orazone> exit

# Step 2: Install the zone (takes several minutes)
zoneadm -z orazone install

# Step 3: Boot the zone
zoneadm -z orazone boot

# Step 4: Initial zone setup (set root password, etc.)
zlogin -C orazone
```

7.4 Zone Cloning — Fast Environment Copy

Zone cloning lets you create a copy of an entire zone (including Oracle installation) in minutes. Perfect for creating DEV/TEST copies of PROD Oracle zones.

```
# Halt the source zone
zoneadm -z orazone halt

# Clone the zone
zoneadm -z orazone_dev clone orazone

# Customize the clone (change hostname, IP)
zonecfg -z orazone_dev
zonecfg:orazone_dev> select net address=192.168.1.100/24
zonecfg:orazone_dev:net> set address=192.168.1.101/24
zonecfg:orazone_dev:net> end
zonecfg:orazone_dev> commit

# Boot both zones
zoneadm -z orazone boot
zoneadm -z orazone_dev boot
```

TIP: ZFS + Zones is a killer combination for Oracle DBAs. Clone the ZFS dataset (instant) + clone the zone = full DEV environment created in minutes. This is your party trick in interviews.

Chapter 8: Solaris Networking for Oracle DBAs

Oracle database connectivity depends on the network — the listener, tnsnames, and client connections all run over TCP/IP. Knowing Solaris networking helps you troubleshoot Oracle connectivity issues.

8.1 Network Configuration — Solaris 10 vs 11

Task	Solaris 10	Solaris 11
List interfaces	ifconfig -a	ipadm show-if
Show IP addresses	ifconfig -a grep inet	ipadm show-addr
Configure IP (temp)	ifconfig e1000g0 192.168.1.10 netmask 255.255.255.0	ipadm create-addr -T static -a 192.168.1.10/24 e1000g0/v4
Bring interface up	ifconfig e1000g0 up	ipadm enable-if -t e1000g0
List physical NICs	dladm show-phys	dladm show-phys
Show link status	dladm show-link	dladm show-link
Default route	route -p add default 192.168.1.1	route -p add default 192.168.1.1
DNS config	vi /etc/resolv.conf	svccfg -s dns/client setprop

8.2 Oracle Listener Troubleshooting on Solaris

```
# Check listener status
lsnrctl status
lsnrctl status LISTENER_ORCL      # named listener

# Check if listener port is open
netstat -an | grep 1521

# Check if listener process is running
ps -ef | grep tnslnsr

# Test connection from client
tnsping ORCL

# Check tnsnames.ora
cat $ORACLE_HOME/network/admin/tnsnames.ora

# Check listener.ora
cat $ORACLE_HOME/network/admin/listener.ora

# Trace Oracle network connection
# Add to sqlnet.ora for tracing
```

```
TRACE_LEVEL_CLIENT = SUPPORT
TRACE_DIRECTORY_CLIENT = /tmp
TRACE_FILE_CLIENT = client_trace
```

8.3 /etc/hosts and Name Resolution

```
# View /etc/hosts
cat /etc/hosts

# Add Oracle server entry
echo '192.168.1.10  oraserver  oraserver.company.com' >> /etc/hosts

# Check name resolution order (/etc/nsswitch.conf)
grep hosts /etc/nsswitch.conf

# Typical Oracle setup - check files first, then DNS
hosts: files dns

# Test name resolution
nslookup oraserver
getent hosts oraserver
```

Chapter 9: Solaris Security for Oracle DBAs

9.1 File Permissions and Oracle

Oracle files have strict permission requirements. Wrong permissions cause Oracle installation failure or security vulnerabilities.

Oracle Directory / File	Owner	Permissions	Why
\$ORACLE_BASE	oracle:oinstall	755	Oracle and oinstall group readable
\$ORACLE_HOME	oracle:oinstall	755	Software home — readable, not world-writable
\$ORACLE_HOME/bin/oracle	oracle:dba	6751	SUID bit — allows DBA group to run oracle binary
\$ORACLE_HOME/bin/sqlplus	oracle:oinstall	755	SQL*Plus binary
Oracle datafiles (.dbf)	oracle:dba	640	Oracle reads/writes, dba group readable
Archive log files	oracle:dba	640	Oracle writes, dba group readable
Trace/alert log files	oracle:dba	644	Readable by all for troubleshooting

```
# Check Oracle binary permissions (must have SUID)
ls -l $ORACLE_HOME/bin/oracle
-rwsr-s--x 1 oracle dba ... oracle  <- correct (s = SUID set)

# Fix Oracle binary permissions if wrong
chmod 6751 $ORACLE_HOME/bin/oracle
chown oracle:dba $ORACLE_HOME/bin/oracle
```

9.2 Solaris Auditing

```
# Check if auditing is enabled
auditconfig -getcond

# View audit classes
auditconfig -lsclass

# Check audit logs
ls /var/audit/
praudit /var/audit/20260420.*      # read audit trail
```

9.3 Transparent Data Encryption (TDE) on Solaris

TDE encrypts Oracle datafiles at rest. On Solaris, the TDE wallet is often stored on a ZFS dataset with restricted permissions.

```
# Create ZFS dataset for TDE wallet
zfs create -o mountpoint=/oracle/wallet orapool/wallet
chmod 700 /oracle/wallet
chown oracle:dba /oracle/wallet

-- Configure wallet in sqlnet.ora
ENCRYPTION_WALLET_LOCATION =
  (SOURCE =
    (METHOD = FILE)
    (METHOD_DATA =
      (DIRECTORY = /oracle/wallet)))

-- Create and open wallet
SQL> ADMINISTER KEY MANAGEMENT CREATE KEYSTORE '/oracle/wallet' IDENTIFIED BY
wallet_password;
SQL> ADMINISTER KEY MANAGEMENT SET KEYSTORE OPEN IDENTIFIED BY wallet_password;
SQL> ADMINISTER KEY MANAGEMENT SET KEY IDENTIFIED BY wallet_password WITH BACKUP;
```

Chapter 10: Solaris Package Management

Before installing Oracle prerequisites on Solaris, you need to install OS packages. Solaris 10 and 11 use completely different package management systems.

10.1 Solaris 10 — SVR4 Packages (pkgadd)

```
# List installed packages
pkginfo                                # all packages
pkginfo -l SUNWxcu4                    # details of a specific package

# Install a package
pkgadd -d /cdrom/sol_10_pkg SUNWxcu4
pkgadd -d /tmp/package.pkg all

# Remove a package
pkgrm SUNWxcu4

# Check if Oracle required packages are installed
pkginfo SUNWarc SUNWbtool SUNWhea SUNWlibm SUNWlibms SUNWsprot SUNWtoo
```

10.2 Solaris 11 — IPS Packages (pkg)

```
# Update package list
pkg refresh

# Search for a package
pkg search oracle
pkg search gcc

# Install a package
pkg install gcc
pkg install system/library/math

# Install Oracle 19c prerequisites (Solaris 11)
pkg install system/library
pkg install system/library/math
pkg install system/library/c++/sunpro
pkg install developer/ assembler

# List installed packages
pkg list
pkg list | grep oracle

# Check package info
```

```
pkg info gcc

# Update all packages
pkg update

# Revert to previous boot environment if update breaks something
beadm list                # list boot environments
beadm activate solaris-backup  # activate previous BE
reboot
```

NOTE: Solaris 11 Boot Environments (BE) are ZFS clones of the OS. Every pkg update creates a new BE automatically. This is a safety net — if an Oracle patch breaks the OS, you can revert in seconds.

Chapter 11: Common Solaris Tasks for Oracle DBAs

These are the tasks you will perform regularly as an Oracle DBA working on Solaris. Practice all of these until they are second nature.

11.1 Finding Files and Oracle Logs

```
# Find Oracle trace files
find $ORACLE_BASE/diag -name '*.trc' -newer /tmp/marker -ls

# Find large files consuming space
find /u01 -size +100M -ls | sort -k7 -rn

# Find Oracle core dump files
find $ORACLE_BASE -name 'core' -ls

# Find files modified in last 24 hours
find /u01/oradata -mtime -1 -ls
```

11.2 Archive Log Space Management

A full archive log destination is one of the most common emergencies for Oracle DBAs on Solaris (ORA-00257).

```
# Check archive log space
df -h /archive

# Find largest archive logs
ls -lSh /archive/ORCL/ | head -20

# Delete old archive logs (always use RMAN — not rm)
RMAN> DELETE ARCHIVELOG ALL COMPLETED BEFORE 'SYSDATE-2';

# Emergency — check ZFS archive dataset
zfs list orapool/arch
zfs get quota,used,available orapool/arch

# Expand ZFS archive dataset quota
zfs set quota=200G orapool/arch
```

11.3 Process Management

```
# Find Oracle processes
ps -ef | grep ora_ | grep -v grep
```



```
# Find a specific Oracle background process
ps -ef | grep pmon

# Get detailed info about a process
pmap -x <PID>                # memory map of process
pstack <PID>                 # stack trace (no recompile needed)
ldd <PID>                     # shared libraries loaded

# Kill a stuck process
kill -9 <PID>

# Find which process is using a file (like a datafile)
lsof /u01/oradata/ORCL/system01.dbf
```

11.4 Cron Jobs for Oracle DBA Automation

```
# Edit oracle user's crontab
crontab -e -u oracle

# Example: RMAN backup every night at 11 PM
0 23 * * * /home/oracle/scripts/rman_backup.sh >> /home/oracle/logs/rman_$(date
+Y%m%d).log 2>&1

# Example: Archive log cleanup every 6 hours
0 */6 * * * /home/oracle/scripts/arch_cleanup.sh

# Example: Tablespace check every 30 minutes
*/30 * * * * /home/oracle/scripts/ts_check.sh

# List current crontab
crontab -l
crontab -l -u oracle

# Cron log on Solaris
tail -f /var/cron/log
```

11.5 Useful One-Liners for Oracle DBAs on Solaris

```
# How many oracle processes are running
ps -ef | grep 'ora_' | grep -v grep | wc -l

# Check SGA size from OS level
ipcs -mb | grep oracle

# Top 10 memory-consuming Oracle processes
prstat -u oracle | head -10
```

```
# Check if Oracle listener is alive
ps -ef | grep tnslnsr | grep -v grep && echo LISTENER UP || echo LISTENER DOWN

# Watch archive log growth in real time
watch -n 30 'ls -lh /archive/ORCL/ | tail -5'

# Count alert log ORA errors today
grep 'ORA-' $ORACLE_BASE/diag/rdbms/orcl/ORCL/trace/alert_ORCL.log | grep $(date +%Y-%m-%d) | wc -l
```

Chapter 12: Interview Preparation — Solaris + Oracle DBA

These are the most commonly asked Solaris questions in Oracle DBA interviews. Know these thoroughly.

12.1 Top Interview Questions & Answers

Q1: What is the difference between ZFS pool and ZFS dataset?

A zpool is a storage pool made of physical disks — it is the container for all storage. A dataset is a logical unit within a zpool — it can be a file system, volume, or snapshot. Think of zpool as an ASM disk group and dataset as an ASM file.

Q2: How do you check if a Solaris server is low on memory?

Use 'vmstat 5' and check the 'free' column (pages free) and 'po' column (pages paged out). If pages are being paged out, memory is under pressure. Also use 'swap -s' to check swap utilization.

Q3: What happens if a Solaris Zone runs out of memory?

If a capped-memory limit is set, Oracle processes inside the zone will get ORA-04031 (out of shared memory) or Oracle will start killing processes. Without a cap, the zone competes with other zones for memory.

Q4: How do you recover a ZFS pool with a failed disk?

```
zpool status                # identify failed disk
zpool replace orapool c0t3d0 c0t5d0  # replace with new disk
zpool status                # monitor resilvering progress
```

Q5: What is the SMF maintenance state and how do you fix it?

The maintenance state means an SMF service has failed and requires manual intervention. You diagnose with 'svcs -x <service>' to read the error, fix the root cause, then run 'svcadm clear <service>' to bring it back online.

Q6: How do you set Oracle kernel parameters on Solaris 11?

On Solaris 11, use projects. Create an Oracle project with resource controls: projadd with project.max-shm-memory, project.max-sem-ids, and process.max-sem-nsems limits set appropriately for the Oracle SGA size.

Q7: What is DTrace and how is it useful for Oracle DBAs?

DTrace is Solaris's dynamic tracing framework. For Oracle DBAs it helps trace system calls made by Oracle processes, identify slow I/O operations, trace network activity, and diagnose issues that Oracle's own tools don't expose — all with zero overhead when not in use.

Q8: How do you find which process is holding a file open on Solaris?

```
fuser /u01/oradata/ORCL/users01.dbf # shows PIDs holding the file
```

Q9: What is ZFS recordsize and why does it matter for Oracle?

ZFS recordsize is the maximum block size ZFS uses for a file. Oracle reads and writes in Oracle block sizes (typically 8KB). If ZFS recordsize doesn't match, ZFS reads larger blocks than needed causing read amplification. Setting 'zfs set recordsize=8k' on the Oracle datafiles dataset ensures optimal I/O alignment.

Q10: How do you grow Oracle storage on Solaris with ZFS without downtime?

```
# Add a new disk to the ZFS pool
zpool add orapool /dev/dsk/c0t8d0
# Increase ZFS dataset quota
zfs set quota=1T orapool/oradata
-- In Oracle: add new datafile or resize existing
SQL> ALTER TABLESPACE DATA ADD DATAFILE '+DATA' SIZE 50G AUTOEXTEND ON;
```

No downtime needed — ZFS and Oracle are both online throughout.

12.2 Skills Matrix — Where You Stand and Where to Go

Skill Area	Your Level	Target Level	Action
OS basics, commands	Intermediate	Strong	Review Chapters 1, 4, 11
File system — UFS	Intermediate	Proficient	Practice Chapter 2.2-2.3
ZFS	Learning	Expert	Deep dive Chapter 2.4 — ZFS is the future
User Management	Intermediate	Strong	Add RBAC — Chapter 3.4
SMF	Beginner	Proficient	Configure Oracle as SMF service
Zones	Beginner	Intermediate	Build a test zone with Oracle inside
Storage — SVM/Veritas	Beginner	Aware	Chapter 6 — interview-level knowledge
Performance monitoring	Intermediate	Strong	Practice prstat, iostat, DTrace
Networking	Basic	Proficient	Chapter 8 for Oracle connectivity
Security / RBAC	Basic	Aware	Chapter 9 key concepts

Your 30-Day Action Plan

Here is a realistic 30-day plan to solidify your Solaris skills before your next project:

Week	Focus	Daily Tasks
Week 1	ZFS Mastery	Set up a Solaris 11 VM. Create zpools, datasets. Practice snapshots, clones. Set Oracle-specific properties (recordsize, sync, compression).
Week 2	Zones + SMF	Create a non-global zone. Install Oracle 19c inside it. Configure Oracle as an SMF service so it auto-starts on reboot.
Week 3	Performance + DTrace	Practice prstat, vmstat, iostat daily. Write 5 DTrace one-liners for Oracle process monitoring. Generate and analyze an AWR report on Solaris.
Week 4	Interview Prep	Answer all 10 interview questions out loud. Build a cheat-sheet of Solaris commands you use daily. Document your project Solaris work in your CV.

Resources to use:

- Oracle Solaris 11.4 — free download at oracle.com/solaris
- Oracle VirtualBox — run Solaris 11 VM for free
- Oracle Docs — docs.oracle.com/cd/E53394_01 (Solaris 11.4)
- Brendan Gregg's DTrace tools — dtrace.org (creator of DTrace tools)
- Oracle My Support — MOS — for Solaris + Oracle integration notes

End of Solaris Administration for Oracle DBA Handbook
Your Solaris + Oracle DBA combination is a rare and valuable skill. Own it.